COMPACT OBJECTS OF THE DERIVED CATEGORY OF QUASI-COHERENT MODULES ON A QUASI-COMPACT QUASI-SEPARATED SCHEME ARE EXACTLY THE PERFECT OBJECTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A perfect object has a dual perfect complex with Hom out of the object computed as cohomology of the tensor product with the dual, and the dual tensor functor commutes with direct sums. invoked at proof lines 4471-4479. Gives the forward implication via the Hom criterion for compactness.
- Global sections over a quasi-compact quasi-separated scheme and the derived pushforward along a quasi-compact separated morphism commute with direct sums in D_QCoh, by the quasi-coherence pushforward direct sums lemma. invoked at proof lines 4480-4482 and proof lines 4491-4493. Used both for the forward implication and for restricting compactness to affine opens.
- The local characterization of perfect objects on affines and the affine comparison with the derived category of modules, together with the More on Algebra proposition that compact objects of D(A) are perfect. invoked at proof lines 4486-4488 and proof lines 4499-4502. Reduces the converse to the affine module case.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cited at proof lines 4472-4473 for the Hom formula; the compactness conclusion is inside the counted span.

Source reference. Original lemma 08DZ: sources/stacks-perfect/pinned-original-source.tex, source lines 803–803. Cited at proof lines 4481 and 4493 for both implications; the direct-sum arguments are inside the counted span.

Source reference. Original lemma 08CN: sources/stacks-chow/tags.txt, source lines 10608–10608. Cited at proof lines 4487-4488 to reduce to affine opens; the adjunction argument is inside the counted span.

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Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 4499-4500 to translate the affine case; the final citation of the module proposition is inside the counted span.

Source reference. Original proposition 07LT: sources/stacks-chow/tags.txt, source lines 9703–9703. Cited at proof lines 4501-4502 to conclude in the affine case; the reduction to it is inside the counted span.

Proposition 0.2. *Let X be a quasi-compact and quasi-separated scheme. An object of $D_{QCoh}(\mathcal{O}_X)$ is compact if and only if it is perfect.*

Proof. If K is a perfect object of $D(\mathcal{O}_X)$ with dual $K^\vee$ (Cohomology, Lemma 08DQ) we have

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K, M) = H^0(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} M)$$

functorially in M. Since $K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} -$ commutes with direct sums and since $H^0(X, -)$ commutes with direct sums on $D_{QCoh}(\mathcal{O}_X)$ by Lemma 08DZ we conclude that K is compact in $D_{QCoh}(\mathcal{O}_X)$.

Conversely, let K be a compact object of $D_{QCoh}(\mathcal{O}_X)$. To show that K is perfect, it suffices to show that $K|_U$ is perfect for every affine open $U \subset X$, see Cohomology, Lemma 08CN. Observe that $j : U \to X$ is a quasi-compact and separated morphism. Hence $Rj_* : D_{QCoh}(\mathcal{O}_U) \to D_{QCoh}(\mathcal{O}_X)$ commutes with direct sums, see Lemma 08DZ. Thus the adjointness of restriction to U and Rj_* implies that $K|_U$ is a compact object of $D_{QCoh}(\mathcal{O}_U)$. Hence we reduce to the case that X is affine.

Assume $X = \mathrm{Spec}(A)$ is affine. By Lemma 06Z0 the problem is translated into the same problem for $D(A)$. For $D(A)$ the result is More on Algebra, Proposition 07LT. □